

1560 Coal bed rock

Department of Mining Engineering
Indian Institute of Technology (ISM), Dhanbad
B. Tech (Open Elective) 6th Semester

Examination: MID SEMESTER

Subject: MNO 401 ROCK ENGINEERING

Instructions: 1. Answer ALL questions
2. Neat & accurate sketches assists you in solving the problems logically and faster
3. Assume logical data, where necessary
4. Use Rock Mechanics Sign Convention to solve the problems

Session: 2024-25

Semester: Winter

Time: 2 HOURS

Max. Marks: 45

Q. No.	Question	Marks
1	(a) Draw neat sketches to derive and discuss the Cauchy's First and Second law of stress. (b) Draw the components of traction acting on an element in two dimensions and prove that $\tau_{xy} = \tau_{yx}$. (c) The rectangular state of stress at a point 'P' in a rock is given by the array $\begin{bmatrix} 15 & 5 & 6 \\ 5 & 5 & 4 \\ 6 & 4 & 10 \end{bmatrix}$ If the normal to a plane makes angles of 30° and 45° with X-axis and Y-axis respectively. Then find the magnitude of the normal stress and the shear stress on the plane.	6 3 6
2	(a) Which global event was responsible for the initiation of an intensive research programme that resulted in major advances in the methods used for designing coal pillars. (b) Write the generalised pillar strength formula given by Salamon and Munro. (c) With reference to the answer of 2(b), what happens to the pillar strength when the pillar height is increased by keeping the pillar width constant. (d) Draw the approximate stress vs strain curve for loading and unloading of a rock specimen. (e) Explain the concept of elastic rebound with respect to the curve drawn in answer to 2(d). (f) What causes elastic rebound in rocks? (g) Differentiate between "Conventional Engineering Design" and "Rock Engineering Design".	1 2 2 1 2 2 5
3	Use compass, set square, scale and pencil to determine the (approximate value of) principal stress for the given stress configuration. [GRAPH PAPER WILL NOT BE SUPPLIED – use the mid semester answerscript to draw the Mohr circle] $\begin{bmatrix} 15 & 3 \\ 3 & 5 \end{bmatrix}$ Determine analytically the principal stresses and directions of the principal stresses to verify the Mohr's Circle solution.	8 7

The relevant formula for solving the problem is given as under.
(The symbols have their usual meaning)

$$(\vec{p}) = [\tau] [\vec{n}],$$

$$\tau = L \tau L^T$$

$$L = \begin{pmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{pmatrix} \text{ for 2D}$$

$$L = \begin{bmatrix} \cos(\angle x'ox) & \cos(\angle x'oy) & \cos(\angle x'oz) \\ \cos(\angle y'ox) & \cos(\angle y'oy) & \cos(\angle y'oz) \\ \cos(\angle z'ox) & \cos(\angle z'oy) & \cos(\angle z'oz) \end{bmatrix} \text{ for 3D}$$